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# SMARTAIR INSIGHTS

www.SmartAirInsights.com



SmartAir Insights

# Our Team



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# Action Plan


01 Project Context

02 Objective Tree

03 Business Understanding

04 Data Understanding

05 Business Model

06 Live Demonstration of Our Website

07 Conclusion




# 1.Project Context

# Problem Statement



01 Lack of real-time pollution data.

02 Rising respiratory diseases.

03 Unknown pollution sources.

04 Difficulty predicting pollution spikes.

# Solution

Predictive AI Models Integration





Scalable and Flexible Architecture



CRISP-DM methodology to to deliver meaningful insights






Intuitive BI Dashboard



Comprehensive Data Insights

# 2.Objective Tree

## A Clear Path to Clean Air

## Air Pollution Goals:

- Reduce air pollution by 10% by 2026 and hospitalizations.
- Analyze, predict, and visualize PM2.5, $O_3$, and other emissions.
- Send automated emails with updated dashboards and notifications every Monday at 10 a.m.

## Objectives:

- Predicting Outdoors Emisions based on their Emissions Values
- Predicting Outdoors Gazes based on their Emissions Values
- Forecast future NO2 emissions based On the Year and The location
- Forecast future PM2.5 concentrations

- **Vehicle Emission Insights:**
  - Visualize VMT trends and compare vehicle emissions by type and region.

  **Objective:**
  - Predicting vehicule Types based on their annual miles traveled

- **Hospitalization ratio:**
  - Build models to predict pollutants and hospitalizations risks.
  - Use charts, maps, and dashboards to highlight trends and patterns in air quality.

  **Objective:**
  - Leveraging Data to Combat Ozone-Related Health Issues

# 3.Business Understanding

## Transforming Data into Strategic Business Value

# Decision-Makers

Public Health Directors

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Transportation Planning Managers

Environmental Compliance Managers

# Sustainable Development Goals (SDGs)




Good Health and Well-being




Sustainable Cities and Communities




Climate Action

# 4.Data Understanding

## Turning Raw Data into Actionable Insights

# 4.Data Comprehension

## Dataset Overview:

- Dataset Name: Air_Quality_VF.csv

## About our dataset:

- gases, vehicle emissions, and hospitalization trends
- Date: Time periods for air quality measurements.
- Location: It specifies Geographic areas.



# 5.Business Model

## Competitive Advantage

# Business *Model Canvas*

## *Key Partners*

- Environmental NGOs
- Public Transit Agencies
- Engineers

## *Key Activities*

- Collect, clean, and integrate Air Quality data.
- Develop and deploy AI models
- Design and maintain Power BI dashboards

## *Key Resources*

- Air Quality from trusted sources
- AI models, Power BI dashboards

## *Value Propositions*

- Optimize traffic and reduce vehicle emissions for cleaner cities.
- Understand air pollution's health impact to improve public well-beingt.
- Monitor pollutant levels to ensure environmental regulation compliance..

## *Customer Relationship*

- Emproving the features based on client's feed back on demand

## *Channel*

- Outreach through companies and organization

## *Customer Segments*

- Public Health Directors
- Environmental Compliance Managers
- Transportation Plannin Managers

## *Cost Structure*

- Research +development costs 300 DT
- Technology +infrastructure costs 500 DT
- Marketing + outreach cost :billboard 1500 DT

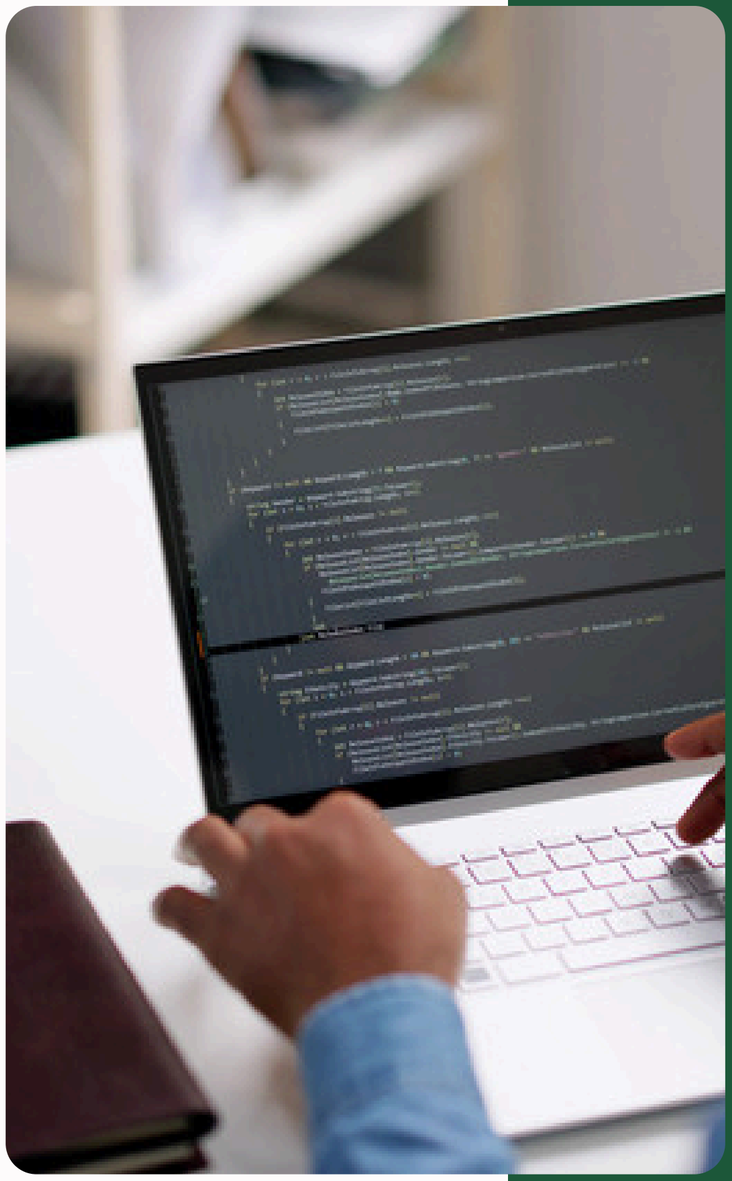
## *Revenue Stream*

- Subscribtion fees 600 DT(per quarter)
- Sepecial offer for the organisational units of the ministry of health 500DT
- Premium extension 750 DT

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# 5.LIVE DEMONSTRATION

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# THANK YOU!


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